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DATA SCIENCE LAB FOR SMART CITIES

FINAL ESSAY

Wildfire Risk and Fuel Management in Portugal: Spatial Coverage Gaps, Economic Vulnerability, and Policy Interventions

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Abstract

Wildfires represent a major environmental and societal challenge in Portugal, with climate change increasing the frequency and severity of fire-prone conditions. Effective prevention requires not only the identification of high-risk areas but also the equitable allocation of protective infrastructure across territories with different economic capacities. This study evaluates how effectively Portugal's Primary Fuel Management Network covers high wildfire risk zones and investigates whether municipalities with lower purchasing power receive proportionate levels of prevention coverage.

Using a Data Science and Smart Cities approach, the analysis integrates wildfire risk data, fuel management infrastructure, municipal boundaries, and purchasing power indicators through Python-based geospatial and statistical methods. Spatial analyses are conducted to assess the alignment between wildfire risk and prevention infrastructure, while municipality-level comparisons explore potential relationships between prevention coverage gaps and economic vulnerability.

Results indicate that X percentage of high-risk areas are covered by the Primary Fuel Management Network. Municipalities with lower purchasing power were found to exhibit lower/higher levels of prevention coverage, suggesting the presence/absence of territorial inequalities in wildfire resilience. These findings highlight the importance of considering both environmental risk and socio-economic conditions when designing prevention strategies and allocating public resources.

This research contributes to the Smart Cities framework by demonstrating how geospatial intelligence and data-driven analysis can support the identification of municipalities where wildfire risk, prevention coverage gaps, and economic vulnerability intersect, enabling more targeted, equitable, and evidence-based policy interventions aimed at improving territorial resilience and climate adaptation.

AI Declaration

This report was developed with the assistance of generative Artificial Intelligence tools, including ChatGPT (OpenAI) and Claude (Anthropic), to support brainstorming, report structuring, methodological planning, literature exploration, writing refinement, and coding guidance.

The authors were responsible for defining the research objectives, selecting and processing the datasets, designing and implementing the analytical methodology, interpreting the results, and formulating the conclusions and policy recommendations. Outputs generated by AI tools were critically evaluated, adapted where necessary, and used solely as supporting resources throughout the development of the project. The authors assume full responsibility for the accuracy, originality, and integrity of the work presented in this report.

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1 Problem Description and Indicators

1.1 Problem Description

Wildfires constitute one of the most significant environmental and territorial challenges affecting Portugal. The country is particularly vulnerable to wildfire events due to a combination of climatic, environmental, and socio-economic factors, including its Mediterranean climate, recurrent drought conditions, extensive forested areas, land abandonment processes, and vegetation accumulation. Climate change has increased wildfire risk across Europe, particularly in Mediterranean regions where prolonged droughts, heatwaves, and extreme fire weather conditions have become more frequent [1]. According to the IPCC Sixth Assessment Report, Southern Europe is expected to experience increasing temperatures, more frequent and intense heatwaves, and more severe drought conditions, all of which contribute to elevated wildfire risk and longer fire seasons [2]. The structural roots of this vulnerability are also socio-economic in nature: since the 1960s, rural exodus and rapid industrialisation left vast inland areas of northern and central Portugal depopulated and effectively unmanaged, driving the expansion of fire-prone shrublands and eucalyptus and pine plantations [3]. Between 1980 and 2023, wildfires in mainland Portugal burned a cumulative total of approximately 4.99 million hectares — an area equivalent to roughly 54% of the national territory [4]. The 2017 fire season alone set an all-time record with approximately 540,000 hectares burned, 114 fatalities, over 1,100 injuries, and economic losses exceeding USD 1.2 billion, making it the costliest natural disaster in the country’s insurance history [5]. More recently, the 2024 fire season burned over 135,000 hectares across central and northern Portugal in a single week [6]. The OECD has estimated that wildfires now affect hundreds of thousands of people annually in Portugal, generate widespread respiratory health impacts, and represent a structural threat to forest ecosystems, soil integrity, and long-term territorial resilience [7].

Prevention, rather than suppression alone, has increasingly been recognised as the most effective strategy for reducing the frequency and severity of wildfires. In this context, Portugal’s Primary Fuel Management Network (*Rede Primária de Faixas de Gestão de Combustível*, RPFGC) constitutes the central instrument of national-scale wildfire prevention. Established under Decree-Law No. 82/2021 and the Integrated Rural Fire Management System (SGIFR), the RPFGC is designed to create large-scale horizontal discontinuities across the landscape through the systematic removal or reduction of vegetation along strategic corridors [8]. Each corridor consists of a central forest road of approximately 6 metres, flanked by 10-metre strips cleared of arboreal and shrub cover, and outer 20-metre strips where canopy spacing is maintained above 4 metres [9]. The network is intended to isolate fire outbreaks, protect roads, critical infrastructure and social facilities, reduce average burned area by compartmentalising the landscape, and improve firefighter safety [10]. Its national implementation is funded with 120 million Euros through the Portuguese Recovery and Resilience Plan (PRR, Project C08-i03), targeting the establishment of 21,000 hectares of fuel management strips across continental Portugal [11]. Despite its strategic importance, the implementation of fuel treatments in Portugal has historically suffered from low execution rates, challenges related to fragmented private land ownership, and insufficient targeting of the territories at highest structural risk [12]. These shortcomings highlight the critical need to evaluate whether the infrastructure being deployed is spatially aligned with zones of greatest wildfire risk, and whether its coverage is equitable across municipalities with differing socio-economic capacities.

This study addresses these challenges by adopting a Smart Cities and Data Science approach to evaluate the spatial effectiveness and equity of Portugal’s primary prevention infrastructure. Smart city frameworks have increasingly been applied to disaster risk management, leveraging geospatial intelligence, remote sensing, and data-driven methods to support more transparent, evidence-based, and equitable public decision-making [13]. In this context, the integration of wildfire risk modelling with infrastructure coverage analysis and socio-economic indicators represents a meaningful contribution to territorial resilience. The central research question guiding this study is: *How effectively does Portugal’s primary fuel management infrastructure cover high wildfire risk zones, are the gaps in prevention coverage disproportionately concentrated in the most economically vulnerable municipalities, and what targeted policy interventions could address these inequalities?* To answer this question, the study pursues three objectives: (1) to assess the spatial alignment between high wildfire risk areas and the coverage of the Primary Fuel Management Network, at the national scale; (2) to analyse coverage gaps at the municipal level and identify the municipalities with the largest proportions of unprotected high-risk territory; and (3) to examine whether municipalities with lower purchasing power exhibit systematically higher prevention coverage gaps, thereby providing evidence of territorial inequalities in wildfire resilience.

1.2 Indicators

To operationalise the research question, four indicators are defined and applied throughout the analysis. Together, they enable a quantitative assessment of wildfire risk exposure, prevention infrastructure coverage, and the socio-economic conditions of municipalities.

Wildfire Risk Index. A composite spatial indicator that quantifies wildfire risk across mainland Portugal on a scale from 1 (very low) to 5 (very high). It is constructed by combining five input layers — conjunctural hazard, historical burned areas, critical fire locations, vegetation index, and aridity index — through a weighted overlay, where each layer is normalised to a common scale prior to combination. The primary fuel management network layer is also sourced from the same institution. For the coverage analysis, only areas classified as risk class 4 (high) or class 5 (very high) are considered.

Coverage Ratio. The proportion of high-risk territory (classes 4–5) that falls within the zone of influence of the Primary Fuel Management Network (RPFGC), defined as a 1 km buffer around the network. A higher value indicates better alignment between prevention infrastructure and high-risk zones.

Gap Ratio. The complement of the Coverage Ratio, quantifying the proportion of high-risk territory that remains outside the buffer and is therefore considered unprotected. It is the primary indicator used in the municipal-level equity analysis to identify territories where prevention coverage is insufficient relative to wildfire exposure.

Purchasing Power. The indicator of economic vulnerability at the municipal level, specifically the per capita income index (IpC), measured relative to the national average (base = 100). The index is inverted and normalised to a 0–1 scale, where 1 represents the most economically vulnerable municipality, and used to examine whether prevention coverage gaps are disproportionately concentrated in economically weaker territories.

Ethical and Social Implications

The indicators selected in this study carry important ethical and social dimensions that extend beyond their technical definition.

The Wildfire Risk Index concentrates the analysis on the territories of greatest fire danger, ensuring that prevention resources and policy attention are directed where they are most needed. The weighting scheme assigns the greatest importance to conjunctural hazard, reflecting the central role of dynamic climatic and meteorological conditions in determining fire danger, while historical burned areas and critical locations capture structural and recurrent risk patterns. The transparency of this construction allows results to be scrutinised and reproduced, which is itself an ethical requirement of evidence-based public policy.

The Coverage and Gap Ratios translate infrastructure presence into a measurable assessment of protected versus unprotected territory. This framing is analytically useful but it is important to acknowledge that proximity to a fuel management corridor does not guarantee complete protection from wildfire. The use of a 1 km buffer is a transparent and operationally justified threshold, supported by sensitivity analysis at 500 m and 2 km, but communities just outside this boundary may still face significant levels of exposure.

The use of purchasing power as a vulnerability indicator reflects a deliberate choice to foreground economic inequality as a lens for evaluating prevention equity. This is ethically significant: it positions wildfire resilience not merely as a technical or environmental problem, but as a social justice issue in which economically marginalised communities, often rural, elderly, and with limited institutional capacity, may bear a disproportionate share of wildfire risk. At the same time, vulnerability is multi-dimensional, and factors such as population ageing, social isolation, and access to emergency services are not fully captured by an economic index alone. Future analyses could benefit from composite vulnerability indicators that integrate these dimensions more comprehensively.

Finally, the municipal scale of analysis is well suited for policy targeting, as it aligns with the administrative units through which prevention investment and governance decisions are made in Portugal. While intra-municipal variation may exist, the municipal level provides the most actionable and institutionally meaningful unit for identifying priority territories and informing resource allocation decisions within the Smart Cities framework.

2 Data Analytics, Optimization and Policy Suggestions

2.1 Datasets

The analysis integrates nine datasets covering wildfire risk, prevention infrastructure, administrative boundaries, and economic vulnerability. Table 1 provides an overview of all datasets used.

Table 1: Datasets used in the analysis

Dataset	Source	Format
Burned Area Perimeters	ICNF	Shapefile
Conjunctural Hazard Maps	ICNF	GeoTIFF
Critical Fire Locations	ICNF	Shapefile
Vegetation Index (IQV)	ICNF	Shapefile
Aridity Index	ICNF	Shapefile
Primary Fuel Management Network	ICNF	Shapefile
Municipal Boundaries (CAOP)	DGT	Shapefile
Portugal Boundaries	ArcGIS Living Atlas	Shapefile
Municipal Purchasing Power (IpC)	INE	CSV

Burned area perimeters record the spatial extent of fire events across mainland Portugal for each year between 2021 and 2024, totalling 5,998 individual fire perimeters across the four-year period. Conjunctural hazard maps provide annual raster layers of fire danger conditions derived from climatic and meteorological variables, available at 25 m resolution for the same period. Critical fire locations identify areas of structural susceptibility based on the *NivelCriti* field. The vegetation index (*Índice de Qualidade da Vegetação*, IQV) and aridity index provide spatial layers of fuel load and drought stress respectively, both at the polygon level. All five datasets, together with the Primary Fuel Management Network layer containing 5,529 network segments, were obtained from the Instituto da Conservação da Natureza e das Florestas (ICNF) [14].

Municipal boundaries were sourced from the Carta Administrativa Oficial de Portugal (CAOP 2024), published by the Direção-Geral do Território (DGT) [15]. This layer provides the official administrative boundaries for all municipalities in mainland Portugal and serves as the spatial unit for the municipal-level analysis.

Purchasing power data were obtained from the INE 2023 Municipal Purchasing Power Study (*Estudo sobre o Poder de Compra Concelhio*), which provides the per capita income index (IpC) for each municipality relative to the national average [16].

2.2 Data Preparation

Data Cleaning
Spatial Harmonization
Municipality-Level Integration

2.3 Wildfire Risk Assessment

Risk Model Construction
Identification of High-Risk Areas

2.4 Prevention Coverage Analysis

Buffer Definition
Coverage Assessment
National Coverage Statistics

2.5 Municipal-Level Analysis

Coverage Ratio by Municipality
Gap Ratio by Municipality
Identification of Priority Municipalities

2.6 Economic Vulnerability Analysis

Purchasing Power Distribution
Economic Vulnerability Mapping

2.7 Relationship Between Coverage Gaps and Economic Vulnerability

Comparative Analysis
Correlation Analysis

2.8 Results and Discussion

Coverage Effectiveness
Economic Vulnerability and Prevention Gaps
Comparison with Existing Literature
Territorial Inequalities in Wildfire Resilience

2.9 Policy Interventions

Priority Municipalities for Intervention
Expansion of the Primary Fuel Management Network
Risk-Informed Resource Allocation
Smart Monitoring and Decision Support Systems

2.10 Ethical and Social Implications of Policy Recommendations

Fairness in Public Investment
Protection of Economically Vulnerable Municipalities
Transparency and Evidence-Based Governance

CRedit Author Statement

*****Author 1: Conceptualization, Methodology, Data Curation, Writing – Original Draft.

Author 2: Data Collection, Data Processing, Visualization, Writing – Review & Editing.

Author 3: Geospatial Analysis, Validation, Investigation, Writing – Review & Editing.

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